

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7				Indicative content: Fission 1. A heavy nucleus captures a slow neutron, splits into lighter nuclei (fission fragments), releases energy and excess neutrons 2. The fission reaction can be controlled safely with boron control rods, which absorb excess neutrons 3. A moderator is needed to slow down the neutrons so they can cause further fission events in a chain reaction Comparison with fusion 4. Two light nuclei collide and fuse producing a heavier nucleus and releasing energy 5. The process requires very high temperature and pressure, making it very difficult to contain. 6. Nuclear fission is splitting of nuclei; fusion is joining of nuclei. 7. Currently fission is used for energy generation but fusion reactors have not yet been built. 8. Badly-controlled fission reactors can overheat and cause meltdown / breach their containment; 9. Fission products are long half-life radioactive materials and require safe storage; Fusion does not produce long half-life radioactive waste. 10. Per kilogram, fusion releases more energy than fission.	6			6		